

Tensor Decomposition for Twitter Social Network Data

Twitter Social Network Data

Twitter social network supplies the use of hashtags which can be viewed as human labelling of the topics in a tweet. A tweet can be responded or passed from user to user by mention and retweet which represent the interactions between users. Monitoring these interactions can reveal some important insights that are going on which is monitoring the conversation between Twitter users.

By monitoring the conversation, there are two important aspects to be concerned. First, the conversation is likely to occur based on social structure. Second, the conversation is evolving over time. This work will consider the first aspect.

The data is collected by using Twitter API. Starting from a seed user, the collection expanded via his follower network until 15,000 users are reached. The collection had been conducted for two months from April 25, 2011 to June 25, 2011. The first month was used as the training set and the evaluation was done on the partial data from the second month.

Tensor Decomposition

A tensor is a generalization of matrix to n dimensions. For example, a cube is a tensor of 3 dimensions or a 3-mode tensor. The index of a tensor can represent a relation, for example the relation (user, movie, rating) can be an index to access the frequency of the relation about movie rating.

In this project, the CANDECOMP/PARAFAC (CP) tensor decomposition is used to identify the top- k clusters of topics and network structures in twitter. CP decompose a tensor X into a low rank approximation of its own Z which can be written by a weighted spectral sum of its factors a, b, c associated with each tensor dimension.

$$X \sim Z = \sum_i \lambda_i \circ a_i \circ b_i \circ c_i$$

The tensor will be constructed from the network structure (focal user, tweet user, action) where action can be either mention or retweet. Another tensor represents the topics of the tweets by (focal user, hashtag) relations. By using tensor decomposition, the resulting weighting factors can be seen as the discovered conversations.

Tensor Representation for Analyzing Twitter Conversation

Consider these relations, Status Relation (statusID, statusText, time_stamp), Mention Relation (statusID, focal_user, tweet_user), Retweet Relation (statusID, focal_user, tweet_user), Hashtag Relation (statusID, hashtag), the objective is to cluster the users based on mention/retweet relations and hashtags/keywords used into communities. Then, the representative hashtags/keywords in the communities will indicate the topic of the conversation. The view for the decomposition joining by statusID becomes (focal_user, tweet_user, hashtag, isMention/Retweet).

Sample Conversation Extracted

The sample conversation is the 11th factor produced from CP decomposition of the tensor of (focal_user, tweet_user, tweet_type (mention/retweet), hashtag).

User	Mentioner/Retweeter	Hashtag	Mention/Retweet
Factor - 11			
0.81 LollyDaskal	0.37 LollyDaskal	0.96 leadfromwithin	0.71/0.7
0.37 JesseLynStoner	0.33 JesseLynStoner	0.25 leadership	
0.24 scedmonds	0.31 scedmonds		
Leadership trainers engaged in motivation related conversation			



Similar to Factor identified before filtering / High Level of Mentioning & Retweeting

Example Tweets

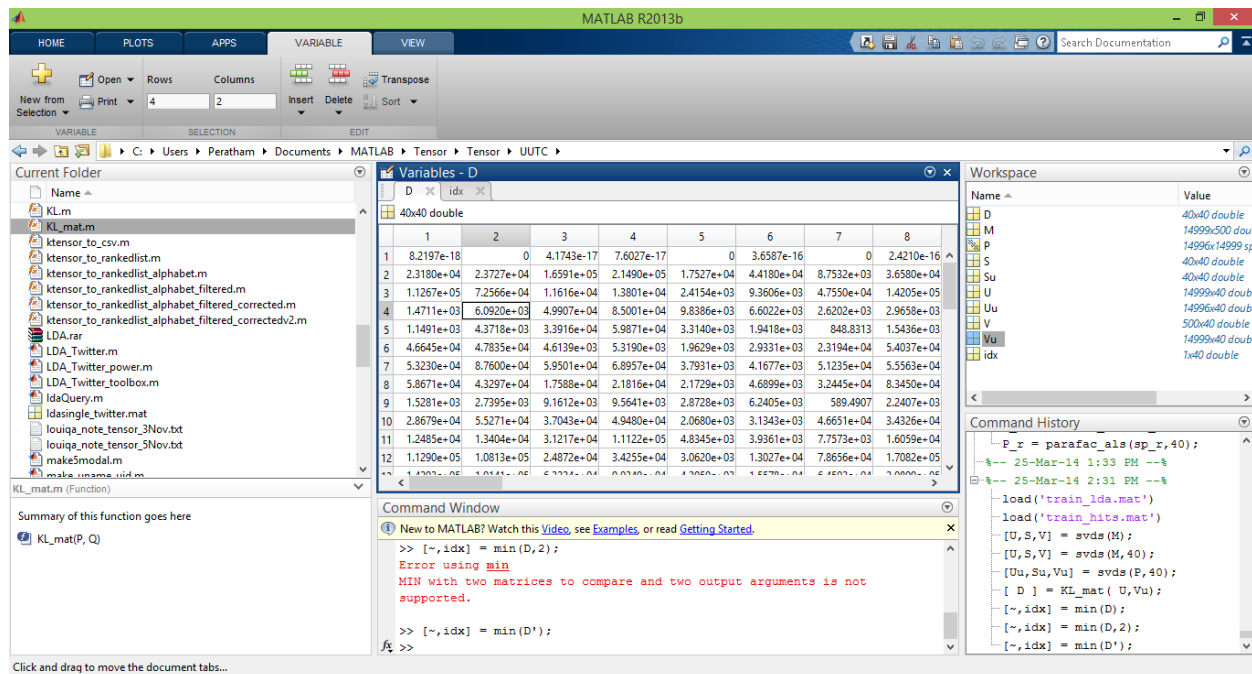
- WELCOME! RT @FearBustinSales: Joining @LollyDaskal in her chat about leadership and humility. #leadfromwithin
- Our thoughts and prayers go out to @scedmonds and his family - Know that we cherish and miss you. #LFW family #leadfromwithin ...

Approximate Tensor Decomposition

Since there are two relations, the framework of *join-by-decomposition* can take advantage of this condition and hopefully provides some speed up over traditional tensor decomposition which restricts with only one relation. In other view, the framework of *decompose-by-normalization* is also a good candidate of automatic functional dependency normalization that may also help speed up the decomposition. The factors from each view are merged by a heuristic based on KL-divergence.

For twitter conversation, I split the view from (focal_user, tweet_user, hashtag, isMention/Retweet) into 2 views (focal_user, tweet_user, isMention/Retweet) and (tweet_user, hashtag) then joining 2 factors together.

For simplicity, I tried a simple case first where I consider (tweet_user, hashtag) and (focal_user, tweet_user). The latter view ignores the relation and treats mentions and retweets as equal. I used SVD on both views and tried to measure KL-divergence for the bases. I got stuck with merging the factors.



Incorporating Keywords

Since hashtags can be seen as user-labelled tags on the tweet, the topical feature could be indicated by just keywords. Here I make a test of correlation between keywords and hashtags by getting a hashtag-keyword co-occurrence matrix and try to see if it is reasonable to use the hashtags as labels.

Temporal Aspect

I will try to create a streaming scenario for the conversation to see the evolving pattern of the topics and communities over time.

References

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